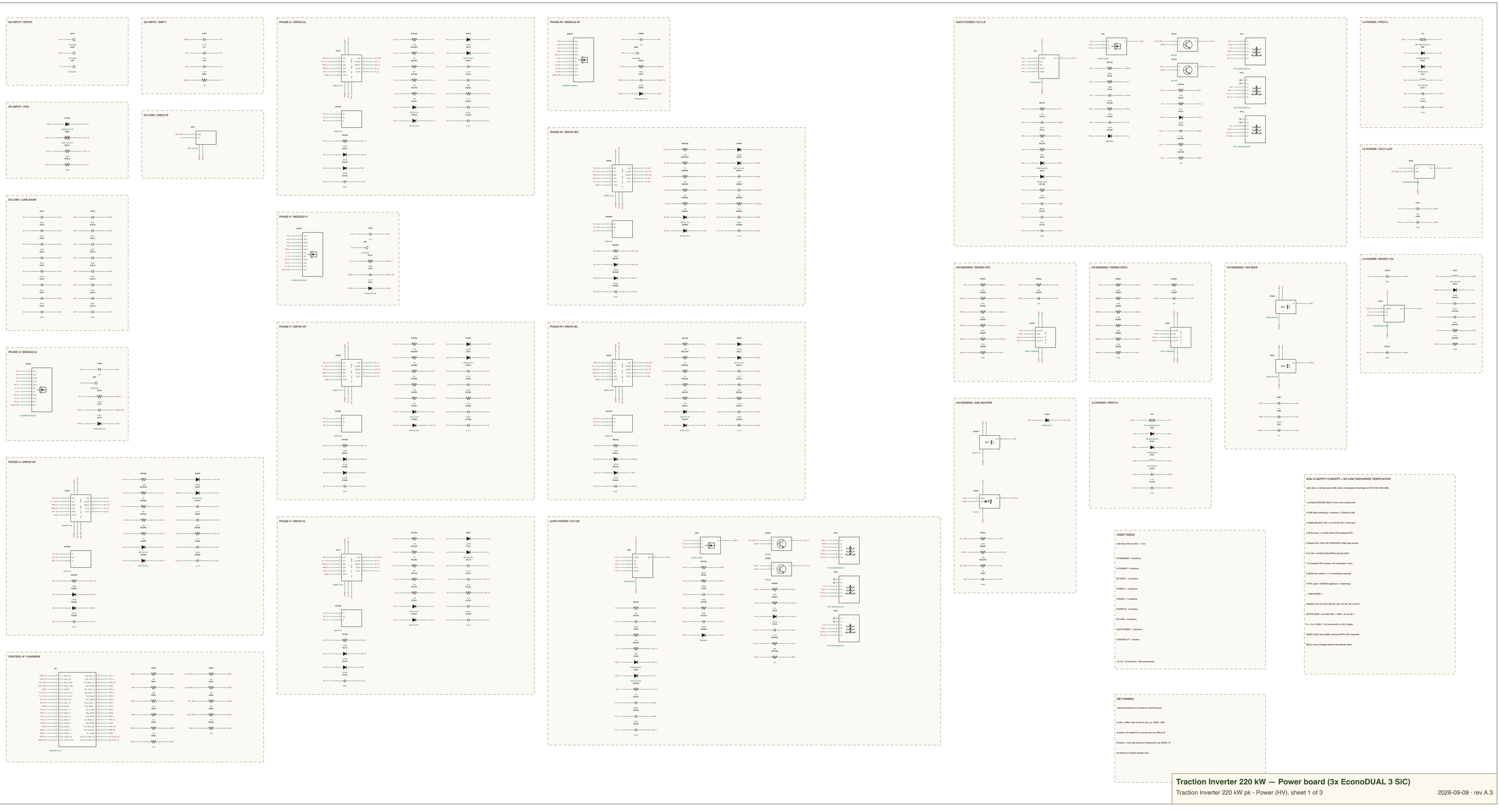
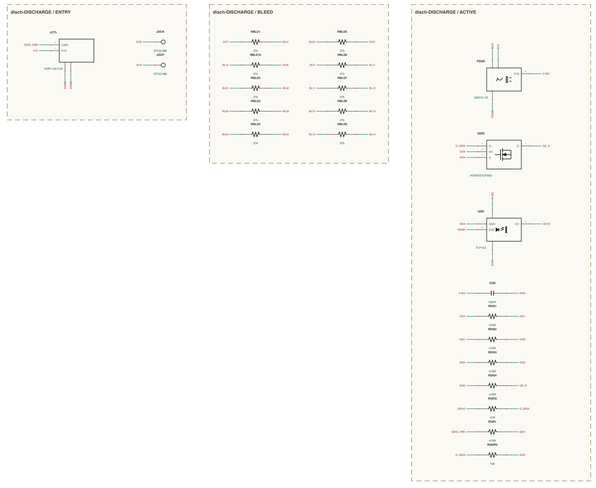

DC-INPUT / ENTRY
DC-INPUT / EMI-Y
DC-INPUT / HVIL
DC-LINK / DISCH-P
DC-LINK / LINK-BANK
PHASE-U / DRIVE-GL
PHASE-W / MODULE-W
PHASE-V / MODULE-V
PHASE-W / DRIVE-WH
PHASE-U / MODULE-U
PHASE-V / DRIVE-VH
PHASE-W / DRIVE-WL
PHASE-U / DRIVE-GH
PHASE-V / DRIVE-VL
GATE-POWER / FLY-HS
CONTROL-IF / HARNESS
GATE-POWER / FLY-LS
HV-SENSING / SENSE-VDC
HV-SENSING / SENSE-VDC2
HV-SENSING / ISO-BIAS
HV-SENSING / ASC-BUFFER
LV-POWER / PROT-H
LV-POWER / PROT-L
LV-POWER / VCC1-LDO
LV-POWER / BOOST-15V
ASIL-D SAFETY CONCEPT + DC-LINK DISCHARGE VERIFICATION
safe state = 3-phase-open (ASC above overspeed); discharge via ECE R100 / ISO 6469
1 backstop SGX3246 (ASIL-D cross) runs torque path
2 FS26 G&A watchdog + monitors + FS0B (ina ENI)
3 FS0B AND MCU. EN = all 6 driver EN + 3-ph open
4 FS1B strap = L3 ASC latch (RE stripped OFF)
5 flyback EN = MCU OR FS26-GPIO1 (ASC gate power)
6 3x half + sum(fed-back) plausibility (torque path)
7 2x isolated VDC sensor, >5% mismatch = fault
8 DESAT per switch, ~1.5 us blanking, latching
9 HVIL open = VDDIO3 signature = discharge
— DISCHARGE —
PASSIVE 10s ETA 2012-DW (5a x 3p x 67.3k): 58 s to 60 V
ACTIVE QSS5 = 4s 470R 10W = 1.88s; 1.6 s to 60 V
E = 116 J 690 V - 33 J(ma worst (u=100 J rated)
VDRP1: MCU ties QSS5, watchdog BOTH VDC channels
RULE: never energize without the bleeder fitted
SHEET INDEX
220 kW pk Power (HV) - 1 of 3
HV-SENSING - 4 sections
LV-POWER - 4 sections
DC-INPUT - 2 sections
PHASE-U - 2 sections
PHASE-V - 2 sections
PHASE-W - 2 sections
DC-LINK - 2 sections
GATE-POWER - 2 sections
CONTROL-IF - 1 section
rev A.3 - 25 sections - 290 components
NET NAMING
internal junctions are named for what they join
Uvsnb_vPHx node at that IC pin e.g. UVSNC_VINP
Rsdemn_W midpoint of a series pair e.g. RBLA_W
Rsdemn_xmtrx tap between Rsdemxmtrx e.g. BVGD_12
all others are explicit design nets
Traction Inverter 220 kW — Power board (3x EconoDUAL 3 SiC)
Traction Inverter 220 kW pk - Power (HV), sheet 1 of 3
2026-09-09 · rev A.3



220 kW pk Discharge (HV, bolt-on) - 2 of 3 - rev A.3 - 3 sections - 24 components

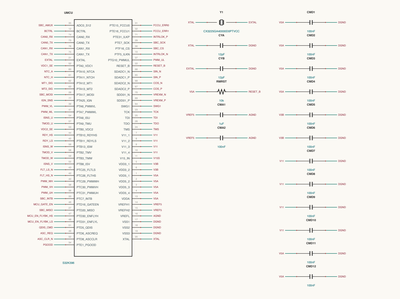
NET NAMING: GndP, -VPhs = nets at that IC pin - Rxxxxx, 2R = series-part footprint - Rxxxxxx_xxxx = tag between Rxxxxxxxxx - all others are explicit design nets

Traction Inverter — Discharge board (bolt-on, XM3 pattern)

Traction Inverter 220 kW pk - Discharge (HV, bolt-on), sheet 2 of 3

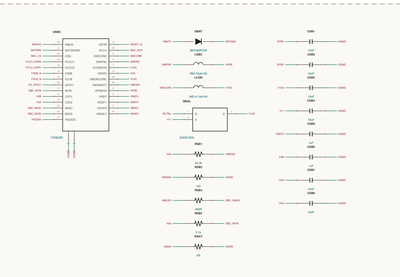
2026-09-09 · rev A.3

CONTROL / MCU

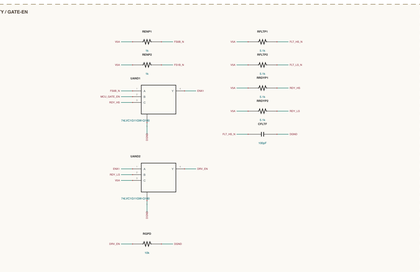


CONTROL / SWD-BOOT

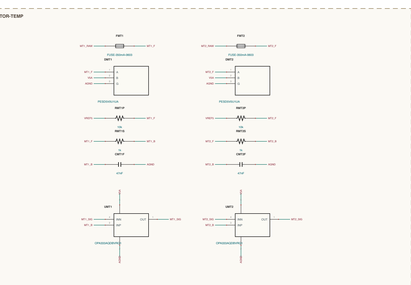
SBC / FS26



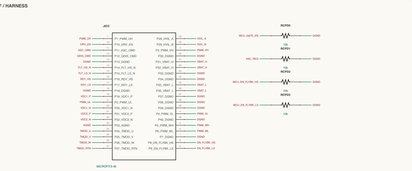
SAFETY / GATE-EN



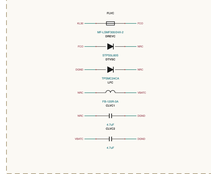
TEMP / MOTOR-TEMP



CARD-IF / HARNESS

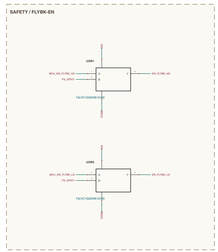


SBC / LV-INPUT

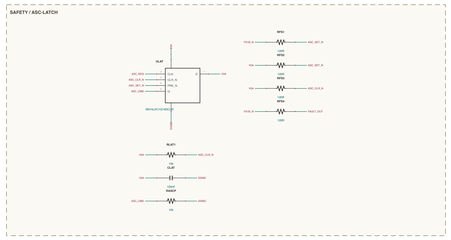


SBC / WAKE

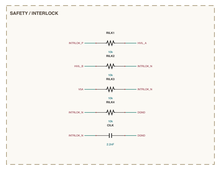
SAFETY / FLYBK-EN



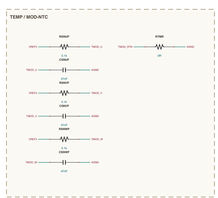
SAFETY / ASC-LATCH



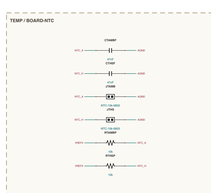
SAFETY / INTERLOCK



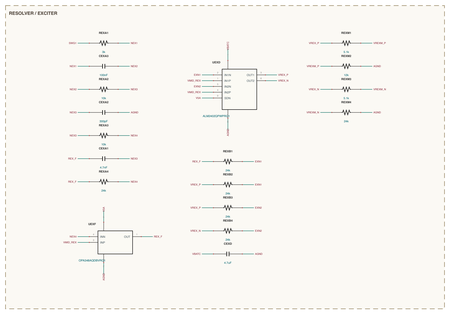
TEMP / MOD-NTC



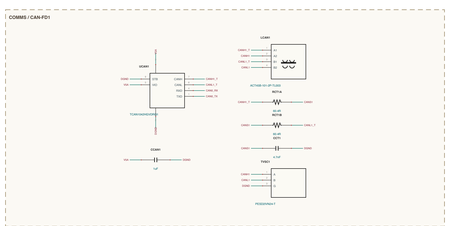
TEMP / BOARD-NTC



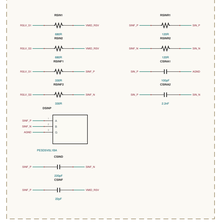
RESOLVER / EXCITER



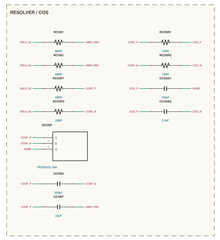
COMMS / CAN-FD1



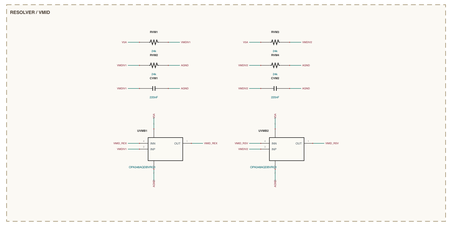
RESOLVER / SIN



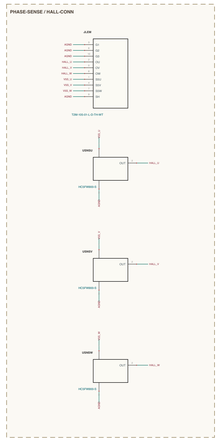
RESOLVER / COS



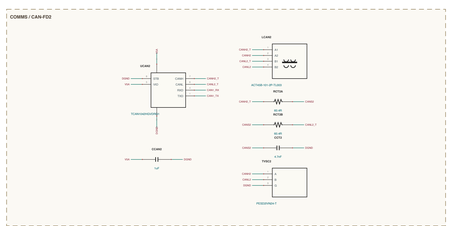
RESOLVER / VMID



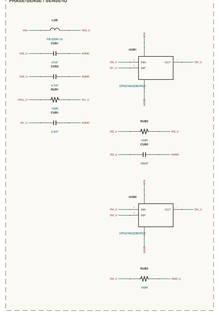
PHASE-SENSE / HALL-CONN



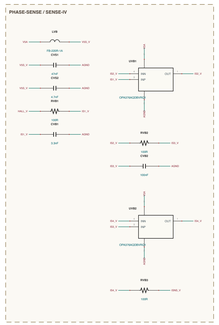
COMMS / CAN-FD2



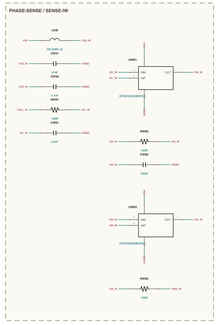
PHASE-SENSE / SENSE-IU



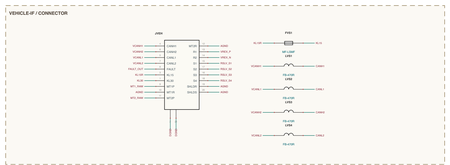
PHASE-SENSE / SENSE-IV



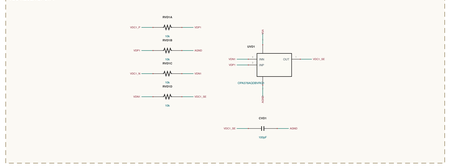
PHASE-SENSE / SENSE-IW



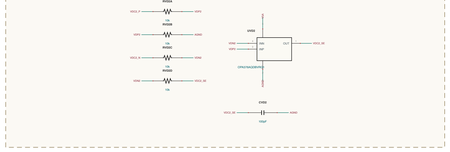
VEHICLE-IF / CONNECTOR



VDC-RECEIVE / CH 1



VDC-RECEIVE / CH 2



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